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Intellectual outcome from 1 to 5 years after epilepsy surgery in 81 children and adolescents: A longitudinal study

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ABSTRACT

Objective: This longitudinal study aimed to measure the time course of intellectual changes after pediatric focal resective epilepsy surgery and to identify their predictors.

Methods: We analyzed a cohort of 81 school-aged children with focal epilepsy and intractable seizures who underwent neurosurgery (focal resection) from 2000 to 2018 in La Timone Hospital (Marseille). Neuropsychological assessments were carried out before and then 1, 2, 3, and 5 years after epilepsy surgery.

Results: Eighty-one patients with a median age at surgery of 13.74 years [4.25] were enrolled. Overall, 45 of the 81 (55%) recruited patients were improved after the surgery on at least one of the five domains of the Wechsler Intelligence Scale. Temporal lobe localization and postoperative seizure freedom were the main prognostic factors impacting intellectual outcome (improvement and decline) after epilepsy surgery. Younger patients at surgery were less likely to have a postoperative IQ decline. Intellectual improvement after epilepsy surgery could be delayed for up to 5 years after surgery and concerned all intellectual domains except the Verbal Comprehension Index (VCI). Intellectual decline after epilepsy surgery occurred mainly during the first two years after the surgery and was reflected in full-scale intelligence quotient (FSIQ) and Working Memory Index (WMI).

Conclusions: Our study points out that children and adolescents with TLE who achieved freedom from seizure after epilepsy surgery are the leading candidates for achieving postoperative intellectual improvement. This enhancement in intellectual function shows a long time course, whereas intellectual decline is evidenced earlier.

Abbreviations

ASM= Anti Seizure Medication
ETLE= Extra Temporal Lobe Epilepsy
FSIQ= Full Scale Intelligence Quotient
IQ= Intelligence Quotient
KM= Kaplan Meier
MEG= Magnetoencephalography

MRI= Magnetic resonance imaging
PET= Positron Emission Tomography
PRI= Perceptual Reasoning Index,
PSI= Processing Speed Index.
WMI= Working Memory Index,
SD= Standard Deviation
SEEG= Stereoelectroencephalography
VCI= Verbal Comprehension Index

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TLE= Temporal Lobe Epilepsy
 VCI= Verbal Comprehension Index
 WAIS-III= Wechsler Adult Intelligence Scale III edition
 WISC IV= Wechsler Intelligence Scale for Children IV edition

1. Introduction

For children with refractory epilepsy, epilepsy surgery is a successful and widely accepted therapeutic option, with reported postoperative seizure freedom rates ranging from 41 to 93%, depending upon the localization and extent of surgery [1, 2]. In operated patients who have achieved seizure freedom, discontinuation of anti-seizure medications (ASM) and improvement of developmental abilities are both ultimate goals.

Cognitive outcomes of epilepsy surgery and their determinants have been primarily studied [3] and a consensus has not yet been reached. Regardless of the cognitive domain studied, stability or improvement has been reported by some authors, while others have noted a decline [4–11]. Numerous preoperative prognosis factors have been identified, such as age at seizure onset, underlying epileptogenic etiology, age at surgery, seizure frequency, duration of epilepsy, and number of ASM trials [8, 9, 11–13], but their respective influence remains difficult to establish [13].

It is challenging to predict postoperative cognitive outcome at the individual level because epilepsy is not a uniform condition but is characterized by clinical, demographic, and etiological heterogeneity. Moreover, additional factors such as ongoing maturational changes, physiological and functional plasticity significantly impact children's cognitive outcome [14, 15]. Following surgery, the reported cognitive function changes are variable. Indeed, there are considerable between-study variations in sample size, etiologies, cognitive domain studied with different psychometric tests [15] and follow-up duration [14, 16–20]. Finally, there are also methodological differences in how reliable/significant cognitive improvements or decline are determined after epilepsy surgery. In this context, there is a need for a longitudinal study with a larger patient cohort, assessed before and after the surgery with standardized tests at fixed time intervals from childhood to early adulthood.

This study aimed to measure intellectual changes, define their time course, and, secondly, identify the predictors of those postoperative cognitive changes after focal resective epilepsy surgery.

2. Material and methods

2.1. Participants

We studied children with refractory focal epilepsy followed in the Marseille tertiary epilepsy center by the multidisciplinary team (including pediatric and adult neurologists, neurosurgeon, and neuropsychologist) at the La Timone and Henri Gastaut hospitals in Marseille, France. We included 81 patients who underwent surgical treatment from 2000 to 2018. Drug resistant epilepsy was defined as failure of adequate trials of two tolerated and appropriately chosen and used AED schedules (whether as monotherapies or in combination) to achieve sustained seizure freedom [21]. The inclusion criteria were: (a) operated before the age of 20 years for epilepsy that started before the age of 16 years (b) having had an extensive evaluation for epilepsy surgery defined as prolonged video-EEG recordings with recorded seizures, interictal and ictal neurologic examination, high-resolution magnetic resonance imaging (MRI) and positron emission tomography (PET) scan and if necessary, depth electrode recordings, functional MRI, magneto-electroencephalography (MEG) (c) multidisciplinary team discussion (d) intellectual skills and behavioral profile of the child allowing reliable testing (e) intellectual evaluation at least one year before surgery (f) at least one year of neuropsychological follow-up after surgery. The preoperative neuropsychological assessment was performed as close as

possible preceding surgery with an interval of no more than one year as recommended by the ILAE Neuropsychology Task Force Diagnostic Methods Commission: 2017–2021 Neuropsychological assessment in epilepsy surgery [22].

We used a truly functional surgical approach, with cortectomy targeting the epileptogenic zone and avoiding resection or disconnection of non-epileptogenic and functional areas. All patients with hippocampal sclerosis underwent an anterior temporal lobectomy.

We excluded patients who underwent extensive (hemispherotomy) or palliative surgery (callosotomy). The children operated at a young age (< 6 yrs), and children who were impaired to a degree that did not allow standardized formal assessment were not considered for the following analysis.

All these cognitive data were collected as part of clinical care then repurposed for research.

2.2. Procedures

2.2.1. Preoperative and postoperative clinical assessments

Each participant was tested individually. All assessments reported in this study were conducted by the same neuropsychologist (V.L.). General intellectual abilities were tested using standardized cognitive assessments at fixed time intervals once a year: within one year before surgery (assessment *pre*) and over the five years following surgery, at one year (*post1*), two years (*post2*), three years (*post3*) and five years (*post5*) post-operatively.

Age-appropriate Wechsler Intelligence Scales (WISC IV or WAIS-III) were used at baseline and follow-up assessments. We report standard scores on five domains: Full-scale IQ (FSIQ), Verbal Comprehension Index (VCI), Perceptual Reasoning Index (PRI), Working Memory Index (WMI), and Processing Speed Index (PSI). The various indices of the Wechsler Intelligence Scale are based on age-corrected data. These standard scores have a mean of 100 and a SD of 15 in the normal population.

Throughout the follow-up period, the same patient may have been tested using two different versions of the Wechsler intelligence scale (WISC IV then WAIS-III). As reported in technical reference manuals of intelligence scales [23, 24], inter-correlations between the different test versions were very high.

2.2.2. Criterion of judgment of postoperative evolution

Postoperative seizure control was classified using Engel's classification criteria [25]. In the postoperative period, the patients were deemed "seizure-free" if they did not have a seizure since the surgery (Engel class Ia).

For each index, a patient was said to be "improved" if the change between pre-and post-surgery was greater than +10 points, "decreased" if the change was greater than -10 points, and "stable" if the change was less than 10 points. We defined a relevant change between pre and postsurgical tests as an increase or decrease of at least 10 points to exclude the test-retest practical effect reported in technical reference manuals of intelligence scales [23, 24].

To identify factors potentially influencing intellectual outcome, we established for each patient a criterion called "intellectual evolution" that summarized the postoperative intellectual outcome. A patient was considered "improved" if at least one index observed a change greater than +10 points, "stable" if all indexes showed changes of less than 10 points, and "decreased" if at least one index decreased with a change greater than -10 points, whatever the evolution on other indexes.

2.3. Statistical analysis

Descriptive statistics were reported as frequencies (sample size and percentages) and medians for categorical and continuous variables, respectively (Table 1–2). The groups were compared using the Mann-Whitney U test and the χ^2 test or Fisher exact test as appropriate. The

Table 1Demographic and clinical characteristics of the epileptic patients according to the epilepsy localization: temporal vs extratemporal^a

	ETLE N= 31	TLE N=50	Total N=81	P-value
Sex (Female)	14 (45)	25 (50)	39 (48)	0.7
Handedness (right)	21 (68)	37 (74)	58 (72)	0.6
Epilepsy characteristics				
Age at epilepsy onset (year)	4.5 (0.1-16)	7 (0.1-16)	6 (0.1-16)	0.001
Epilepsy localization (Right Left)				
Frontal	9 (7 2) (29)			
Parieto occipital	7 (3 4) (23)			
Insulo-perisylvian	15 (12 3) (48)			
Epilepsy laterality (Right/Left)	22 (71) 9 (29)	27 (54) 25 (50)	49 (60) 25 (42)	0.1
Abnormal brain MRI	19 (61)	44 (88)	63 (78)	0.005
Ganglioglioma DNET	2 (6) 3 (10)	8 (16) 15 (30)	10 (12) 18 (22)	ns 0.03
Focal cortical dysplasia	10 (32)	6 (12)	16 (20)	0.03
Perinatal or post-natal stroke	4 (13)	3 (6)	7 (9)	0.3
Hippocampal sclerosis ^b	0	12 (24)	12 (15)	0.003
SEEG	28 (90)	24 (48)	52 (64)	<0.001
ASM before the epilepsy surgery				
Number of	3 (1-4)	1 (1-3)	2 (1-4)	<0.001
Monotherapy	6 (19)	27 (54)	33 (41)	0.002
Baseline intellectual assessment				
Age at presurgical intellectual assessment (year)	13 (6-19)	13.5 (6-19)	13 (6-19)	0.5
FSIQ	81 (± 17)	84 (±19)	83 (± 17.7)	0.5
VCI	88 (±16.0)	85 (±18.7)	88 (± 17.6)	0.9
PRI	80 (±16)	86 (±18.5)	84 (±17.7)	0.2
WMI	84 (±18.4)	85 (±18.6)	85 (±17.4)	0.8
PSI	82 (±15.8)	86 (±17.8)	84 (± 17.1)	0.3
Epilepsy surgery				
Time to epilepsy surgery	8.5 (2-18.9)	6 (1-14)	7 (1-18.9)	0.02
Age at surgery (year)	14 (7-20)	14.6 (6.5-20)	14 (6.5-20)	0.4

ASM: Anti-Seizure Medication; DNET Dysembryoplastic NeuroEpithelial Tumor, ETLE: Extra temporal Lobe Epilepsy; FSIQ Full Scale Intellectual Quotient, MRI Magnetic Resonance Imaging, PRI Perceptual Reasoning Index, PSI Processing Speed Index, SEEG: stereoelectroencephalography TLE Temporal Lobe Epilepsy, VCI Verbal Comprehension Index, WMI Working Memory Index.

^a Data are presented as N (%) or median (interquartile range) or mean (±standard deviation)

^b All patients with hippocampal sclerosis underwent an anterior temporal lobectomy

Table 2Demographic and clinical characteristics of the epileptic patients according to the seizure freedom after their epilepsy surgery^a

	Not seizure Free N= 24	Seizure free N=57	Total N=81	P-value
Gender (Female)	13 (54)	26 (46)	39 (48)	0.7
Handedness (right)	18 (75)	42 (74)	60 (74)	0.9
Epilepsy characteristics				
Age at epilepsy onset (year)	4.3 (0.1-16)	7.0 (0.1-16)	6 (0.1-16)	0.07
Epilepsy location (TLE ETLE)	11(46) 13 (54)	39 (70) 18 (30)	50 (62) 31 (38)	0.06
Epilepsy laterality Right/Left	15 (62) 9 (38)	34 (60) 23 (40)	49 (60) 32 (40)	0.06
Abnormal brain MRI	14 (58)	49 (86)	63 (78)	0.006*
Ganglioglioma DNET	1 (4) 4 (17)	6 (10) 13 (23)	7 (9) 17 (21)	0.35 0.5
Focal cortical dysplasia	6 (25)	10 (17)	16 (28)	0.4
Perinatal or post-natal stroke	0	7 (12)	7 (9)	0.2
Hippocampal sclerosis	1 (4)	11 (19)	12 (15)	0.08
SEEG	14 (58)	37 (65)	51 (63)	0.8
ASM before the epilepsy surgery				
Number of	2 (1-4)	1 (1-4)	2 (1-4)	0.009*
Monotherapy	3 (12)	30 (53)	33 (41)	<0.001
Age at presurgical intellectual assessment (year)	15.3 (8-19)	15 (7.5-19)	15 (7.5-19)	0.9
Epilepsy surgery				
Time to epilepsy surgery	8.75 (1-19)	7 (1-19)	7 (1-19)	0.2
Age at surgery (year)	14.3 (7-19)	14 (6.5-19)	14 (6.5-19)	0.8
Baseline intellectual assessment				
FSIQ	80.5 (±18.1)	84.2 (±18.5)	84.8 (±16.7)	0.47
VCI	86.3 (±17.3)	89.3 (±17.8)	88.4 (±17.6)	0.7
RPI	81.0 (±20.0)	84.8 (±16.7)	83.7 (±17.7)	0.3
WMI	81 (±16.5)	86.1 (±19.1)	85 (±18.4)	0.4
PSI	78.7 (±17.1)	86.6 (±16.7)	84.2 (±17.1)	0.04

ASM: Anti Seizure Medication, DNET Dysembryoplastic Neuroepithelial tumor, ETLE: Extra temporal Lobe Epilepsy; FSIQ Full Scale Intellectual Quotient, MRI Magnetic Resonance Imaging, TLE Temporal Lobe Epilepsy, SEEG: stereoelectroencephalography.

^a Data are presented as N (%) or mean (± Standard Deviation)

baseline scores of all five domains of Wechsler scale were compared to the norm with t test of Student.

Kaplan-Meier survival analyses were used to investigate the difference in time to the emergence of intellectual improvement or decline between the seizure outcome groups (seizure-free vs. not seizure-free), defined as the time elapsed between the epilepsy surgery to the neurocognitive assessment. Patients were censored from this analysis at their last follow-up if they had not reached such intellectual improvement or decline. Statistical significance was tested using the log-rank test and comparison of 95% confidence intervals (CIs).

Binomial logistic regression was performed to ascertain the predicting factors of intellectual changes after the epilepsy surgery. The potential predictors of intellectual outcome were sex; age at epilepsy onset; age at surgery; epilepsy localization; epilepsy lateralization; baseline FSIQ; brain MRI (normal/ abnormal); use of SEEG; time to surgery from the epilepsy onset; epilepsy outcome (seizure-free vs. active epilepsy); number of ASM and decrease of ASM.

The linearity of the continuous variables with respect to the logit of the dependent variable was assessed via the Box-Tidwell procedure. A Bonferroni correction was applied using all fourteen terms in the model, resulting in statistical significance being accepted when $p < 0.003$. Based on this assessment, all continuous independent variables were found to be linearly related to the logit of the dependent variable. There was no multicollinearity between categorical dependant variables. Correlations were assessed by the Spearman non-parametric test.

All analyzes were performed using IBM SPSS Statistics 20.0 (IBM Inc., New York, USA). For all two-tailed tests used, a significance level of $p < 0.05$ was considered statistically significant.

3. Results

3.1. Patient characteristics (Table 1)

We recruited 81 patients: 50 patients with temporal lobe epilepsy (TLE) and 31 with extratemporal lobe epilepsy (ETLE). The TLE group had later onset of epilepsy than the ETLE group (7 vs. 4 years, $p=0.001$); in addition, the TLE group more often had an abnormal MRI (88% vs. 61%, $p=0.005$), had higher numbers of patients on monotherapy (54% vs. 19%, $p=0.002$) and tended to be operated earlier than the ETLE group (6 vs. 8.5 years, $p=0.02$). The TLE group less often had depth electrode recordings (48% vs. 90%, $p=0.001$). At baseline, the patients with TLE, compared to patients with ETLE, showed no difference in term of IQ scores in all intellectual domains (Table 1). No other significant differences were found (handedness, sex, age at presurgical assessment, age at surgery).

3.2. Follow up (Fig. 1)

Five patients were lost to neurological and neuropsychological follow up from *post3*, but were all noted to be seizure-free at their last assessment. Since the study is longitudinal, 8 patients have not yet arrived at *post5*.

3.3. Seizure outcome (Table 2)

One year after surgery, seizure freedom rates (Engel class I) ranged from 78% in children with TLE to 58% in children with ETLE ($p=0.06$). This percentage remained stable over the five years of the follow-up; seizure relapse was observed in only one patient. Children with abnormal brain MRI more frequently became seizure-free (86% vs. 58%, $p=0.006$).

3.4. Neuropsychological assessment

3.4.1. Impact of epilepsy factors on baseline intellectual abilities

Concerning preoperative IQ, in all five domains, patients had significant lower mean scores than the general population (standard scores having a mean of 100 and an SD of 15 from the norm, p value < 0.001 for the comparison of all five domains to the norm), (Table 1).

The patients who started their epilepsy at an older age had a higher preoperative FSIQ ($r=0.24$; $p=0.01$), were more prone to have TLE ($r=0.36$, $p=0.012$), had fewer ASM ($r=0.25$; $p=0.03$), and a shorter time to surgery ($r=0.49$; $p=0.02$). Epilepsy lateralization and abnormal MRI (mean 82.13 vs. 86.65, $p=0.36$) did not influence the preoperative FSIQ.

Among the 81 recruited children, 29 underwent two presurgical intellectual evaluations: one performed within one year of surgery, and the other completed in the 2 to 3 years before surgery. Of these multi-tested children, 16 had ETLE and 13 TLE. We showed no significant difference between the clinical characteristics of children tested twice before surgery and those of children tested once. Of the 29 patients tested twice before the surgery: 2 of 29 (7%) improved on at least one of the five intellectual domains, 11 (38%) were stable, and 16 (55 %) declined, regardless of the epilepsy lateralization, localization and postoperative seizure outcome. Decrease is observed on standard scores. No patient showed a decrease in their raw scores.

3.4.2. Intellectual outcome after surgery. (Table 3)

Overall, after the surgery, 45 of the 81 (55%) recruited patients improved on at least one of the five intellectual domains. Of the improved patients, 35/45 (78%) had TLE, and 37/45 (82%) were seizure-free. Conversely, 27 patients (33%) were impaired, of which 17/27 (63%) had ETLE, and 12/27 (57%) were not seizure-free. The patients who experienced a decline in one index showed a stability or a decline for the other indexes of the Wechsler scale. The patients with TLE were more frequently improved in terms of intellectual outcome ($p=0.002$). Furthermore, of 10/81 (12%) patients who did not experience any intellectual change, 5/10 (50%) had TLE, and 7/10 (70%) were seizure-free after the surgery.

- Neurocognitive improvement: predicting factors and time course

With univariate analysis (Table 4), temporal lobe localization of epilepsy, number of ASM, and seizure-free state were candidate predicting factors (Table 3 A). Multivariate analysis (Table 4 B) revealed the odds of achieving intellectual improvement increased by a factor of 3.5 (95% confidence interval [CI] 1.2-10.6) for seizure-free patients. Furthermore, TLE localization increased the odds of achieving

Table 3

IQ index scores (and SDs) for the seizure-free and non-seizure groups before surgery and for each year of follow-up.

	Baseline Seizure free	Not Seizure free	P value	Post1 Seizure free	Not Seizure free	P value	Post 2 Seizure free	Not Seizure free	P value	Post3 Seizure free	Not Seizure free	P value	Post 5 Seizure free	Not Seizure free	P value
FSIQ	81 ± 17	84 ± 19	0.5	87±17.3	78±16.3	0.04	88±19.3	77±17.3	0.02	88±19.8	75±23	0.03	89±19.9	78±18.1	0.04
VCI	88 ± 16	85 ± 18.7	0.9	91±18.7	83±13.8	0.05	90±19.1	82±14.3	0.05	91±20.9	82±15.2	0.03	91±20.8	80±15.5	0.04
PRI	80 ± 16	86 ± 18.5	0.2	88±18.2	82±19.9	0.04	89±18.3	81±19.3	0.04	89±18.6	81±18.7	0.04	89±18.1	82±19.7	0.04
WMI	84 ± 18.4	85 ± 18.6	0.8	88±17	77±16.7	0.02	89±16.7	78 ± 16.5)	0.02	88±15.4	79±17.2	0.04	89±16.4	79±17	0.003
PSI	82 ± 15.8	86 ± 17.8	0.3	90±18.4	79±17.6	0.01	93±17.9	80±15.6	0.01	91±19	81±14.5	0.02	92±18.8	82±14.6	0.02

Table 4
Predicting factors of Improvement (A-B) and decline (C-D).

A. Univariate analysis, Hazard ratios, 95% Confidence Interval (CI), and p values for predictors of improvement				
Predicting factor	Global P-value	Hazard Ratio (HR)	95% CI for HR Lower Upper	
Sex	0.55	1.3	0.5	3.1
Age at epilepsy onset	0.28	0.9	0.8	1.1
Epilepsy Location: TLE	0.001	5.2	1.9	13.9
Epilepsy Laterality (Right/Left)	0.7	1.2	0.5	2.9
Abnormal brain MRI	0.4	1.6	0.5	4.7
Age at epilepsy surgery	0.8	0.9	0.9	1.1
Time to surgery	0.4	0.9	0.9	1.1
ASM number	0.04	2.0	1.2	3.4
ASM decrease	0.4	0.6	0.2	1.9
Seizure freedom after surgery	0.004	4.4	1.6	12.0
Baseline FSIQ	0.242	1.03	0.9	1.0
B. Multivariate analysis adjusted Odds Ratios, 95% Confidence Interval (CI), and P values for predictors of improvement				
Predicting factor	Global	Adjusted Odds Ratio (OR)	95% CI for OR Lower Upper	
Epilepsy Localization: TLE	0.01	4.2	1.4	12.4
Seizure freedom after surgery	0.02	3.5	1.2	10.6
C. Univariate analysis, Hazard ratios, 95% Confidence Interval (CI), and p values for predictors of decline				
Predicting factor	P-value	Hazard Ratio (HR)	95% CI for HR Lower Upper	
Sex	0.4	1.5	0.6	3.5
Age at epilepsy onset	0.056	0.9	0.8	1.0
Epilepsy Localization: TLE	0.003	0.23	0.09	0.6
Epilepsy Laterality	0.3	0.6	0.2	1.6
Abnormal brain MRI	0.09	0.4	0.1	1.1
Age at epilepsy surgery	0.01	0.9	0.8	0.9
Time to surgery	0.5	0.9	0.9	1.1
ASM Number	0.9	1.0	0.6	1.6
Seizure freedom after surgery	0.007	4.0	1.4	11
Baseline FSIQ	0.4	0.43	0.9	1.0
D. Multivariate analysis adjusted Odds Ratios, 95% Confidence Interval (CI), and P values for predictors of decline				
Predicting factor	P-Value	Adjusted Odds Ratio (OR)	95% CI for OR Lower Upper	
Age at epilepsy surgery	0.02	0.8	0.7	0.9
Epilepsy Location: ETLE	0.02	3.9	1.2	12.3
Active epilepsy after surgery	0.03	3.8	1.2	12.5

ASM: Anti Seizure Medication; ETLE: Extra temporal Lobe Epilepsy; FSIQ Full Scale Intellectual Quotient, MRI Magnetic Resonance Imaging, TLE Temporal Lobe Epilepsy.

intellectual improvement by a factor of 4.2 (95% confidence interval [CI] 1.4-12.4). The model explained 31.3% (Nagelkerke R^2) of the variance in intellectual improvement and correctly classified 72.8% of cases. There was no multicollinearity between epilepsy localization and seizure outcome.

During the five years of follow-up, the patients deemed seizure-free after surgery were more likely to experience intellectual improvement than patients with active epilepsy ($p<0.001$).

The median time to observation of intellectual improvement was 1.3 years (IQR 1.1-2.5) for patients with TLE vs. 1.6. years (IQR 1.2-1.7) for those with ETLE ($p=0.7$). The neurocognitive improvement concerned

different intellectual domains, notably FSIQ (log rank= 0.01), PRI (log rank=0.02), WMI (log rank=0.02) and PSI (log rank=0.037) but not the VCI (Fig. 2).

We observed seizure free patients' continued improvement through all five years of follow-up from 1 to 5 years (Fig. 2.A, Table 5A).

- Neurocognitive decline: predicting factors and time course

With univariate analysis, age at epilepsy onset and surgery, ETLE localization, normal brain MRI, seizure freedom and number of ASM appeared to be candidate predicting factors (Table 4 C). Multivariate analysis (Table 4 D) revealed that the odds of exhibiting intellectual decline decreased by a factor of 0.8 (95% confidence interval [CI] 0.7-0.9) for patients who were younger at the time of epilepsy surgery. Furthermore, ETLE localization and active epilepsy both increased the risk of experiencing intellectual decline by a factor of 3.9 (95% confidence interval [CI] 1.2-12.3) and 95% confidence interval [CI] 1.2-12.5), respectively). The model explained 32.1% (Nagelkerke R^2) of the variance in intellectual decline and correctly classified 72.8% of cases. There was no multicollinearity between epilepsy localization and seizure outcome.

Throughout the five years of follow-up, the patients with active epilepsy after surgery were more prone to experience intellectual decline than seizure-free patients ($p<0.001$). Median time of intellectual decline occurrence was 1.4 years (IQR 1.2-1.8) for patients with TLE vs. 1.6 years (IQR 1.1-1.9) for those with ETLE ($p=0.8$). Neurocognitive decline concerned the FSIQ (log rank= 0.01) and the WMI (log rank=0.05) (Fig. 3). We observed patients' decline throughout the first 2 years of follow-up (Fig. 3, Table 5.B.).

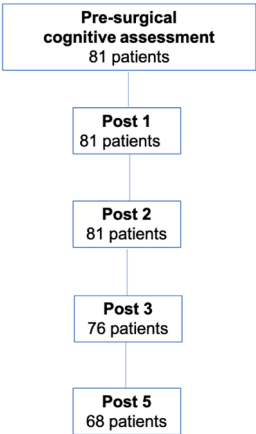


Fig. 1. Flow chart of intellectual assessment during the 5 years of follow-up.

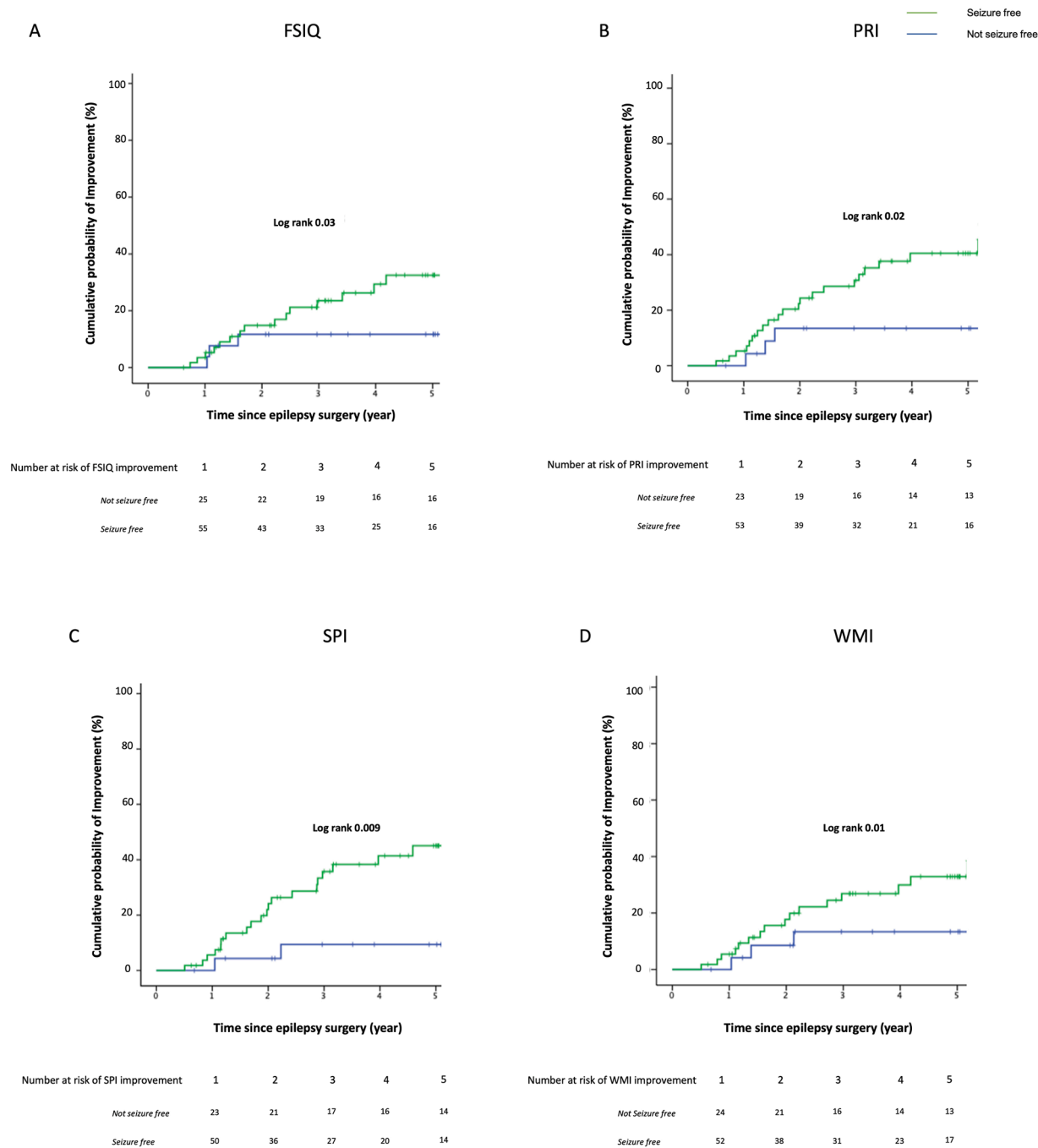


Fig. 2. Kaplan-Meier survival curves showing the cumulative probability of Improvement of Wechsler score index (A-FSIQ, B-PRI, C-PSI, D-WMI) in epileptic patients after surgery, seizure-free *versus* not seizure-free. The numbers of patients remaining at probability of improvement are given alongside the X-axis, denoting the time in years. Tick marks indicate censored patients at the point of their last known follow-up. FSIQ: Full-Scale Intelligence Quotient; PRI: Perceptual Reasoning Index, WMI : Working Memory Index, PSI: Processing Speed Index.

Table 5

Intellectual improvement (A) and decline (B) for the seizure-free and non-seizure free groups for each year of follow-up after epilepsy surgery. ^{a,b,c,d}

A.	Post 1	Post 2	Post 3	Post 4	Post 5
FSIQ					
Not seizure free	4 ± 4	12 ± 6	-	-	-
Seizure free	4 ± 2	15 ± 4	24 ± 6	30 ± 6	33 ± 7
WMI					
Not seizure free	4 ± 4	13 ± 7	-	-	-
Seizure free	5 ± 3	18 ± 5	27 ± 6	30 ± 6	38 ± 7
PRI					
Not seizure free	4 ± 4	13 ± 7	-	-	-
Seizure free	5 ± 3	22 ± 6	31 ± 6	40 ± 6	45 ± 8
PSI					
Not seizure free	4 ± 7	9 ± 6	-	-	-
Seizure free	6 ± 3	22 ± 6	36 ± 7	41 ± 7	45 ± 8

B.	Post 1	Post 2	Post 3	Post 4	Post 5
FSIQ					
Not seizure free	8 ± 6	25 ± 9	-	-	-
Seizure free	2 ± 2	7 ± 4	-	-	-
WMI					
Not seizure free	8 ± 6	25 ± 9	-	-	-
Seizure free	3 ± 2	13 ± 4	-	-	19 ± 7%

^a Data are presented as percentage of improved patients ± Confidence Interval %,
^b (-) For the “not seizure free group”, no more improvement was observed after at 1 and 2 years respectively.
^c Data are presented as percentage % of declined patients ± Confidence Interval %,
^d (-) no more decline was observed

4. Discussion

The present study evaluated the pre- and postoperative intellectual performance in 81 consecutive school-aged children with intractable epilepsy who underwent a focal resection. Our results provide a global picture of intellectual changes over the five years after pediatric epilepsy surgery.

4.1. Seizure freedom rates

Over the five years of follow-up, 78% of children with temporal lobectomy and 58% of children with extratemporal resections achieved freedom from seizures. Those rates are similar to other studies focusing on curative surgery [16, 26–28] [29], with follow-up duration ranging from one year [30–32] to long-term [8, 26, 33]. After surgery, seizure outcome can be heterogeneous in the literature and depends on reported population characteristics. For example, Helmstaedter et al. [5] observed a global seizure-free rate of 80% against 70% in our study. This difference could be explained by an increased percentage of those with normal brain MRI, a large proportion of cortical dysplasia, no brain tumors, and exclusion of patients with hemispherotomy in our study.

4.2. Neuropsychological data

4.2.1. Before surgery

Our results confirm that children with epilepsy generally do not have normal neurodevelopment. Most of them already have deficits in FSIQ and one or more intellectual domains before surgery [5, 15, 34, 35] due to the impact of epilepsy and its medication on brain maturation and development.

Here, we highlight those factors such as earlier age at epilepsy onset, long delay until surgery and a high number of ASM were associated with poorer baseline intellectual abilities. In contrast, the presence or absence of structural abnormalities on cerebral MRI and age at surgery were not predictive of FSIQ before epilepsy surgery. Furthermore, epilepsy lateralization did not influence the baseline intellectual functions, indicating that the epilepsy impact on brain maturation and consequent global intellectual impairment appeared non-specific in our population.

Similar results were demonstrated previously for extratemporal [7, 36] and temporal epilepsy [37, 38]. Furthermore, in the present series, more than half (55%) of patients tested twice before the surgery showed an intellectual decline. Indeed, a lack of typical practice effect was observed in adults with temporal lobe epilepsy compared to controls [39, 40] and children with epilepsy achieved lower test scores during repeat testing on some cognitive measures [27, 28].

Many authors suggest that baseline cognitive performance reflects

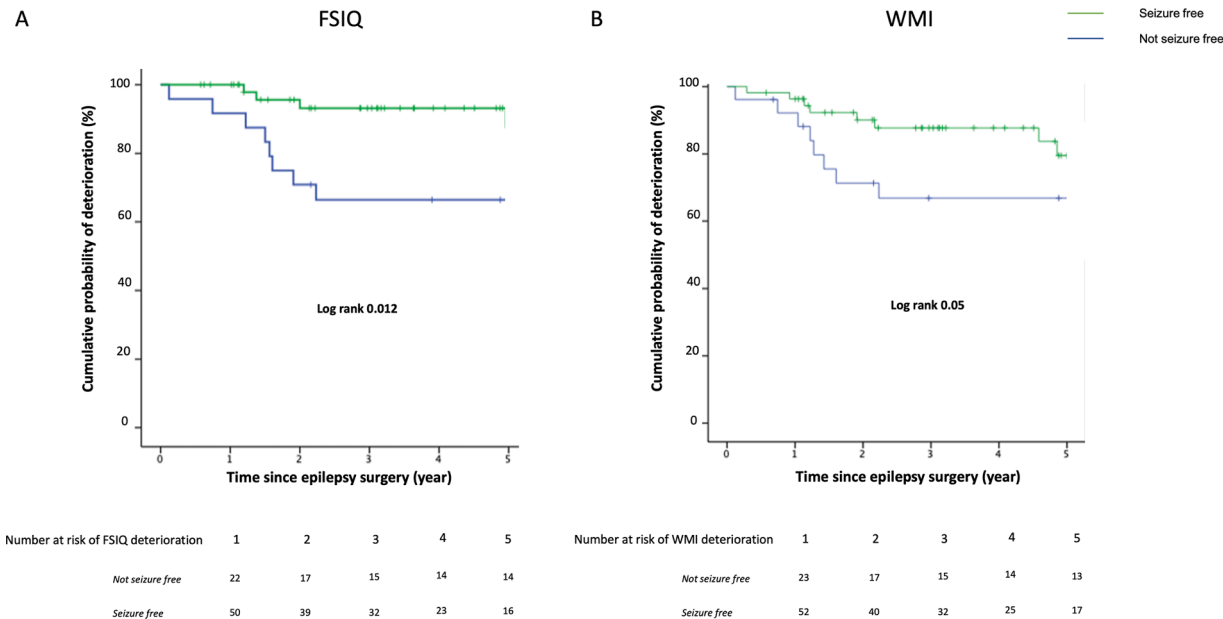


Fig. 3. Kaplan-Meier survival curves showing the cumulative probability of deterioration of Wechsler score index (A-FSIQ, B-WMI) in epileptic patients after surgery, seizure-free versus not seizure-free. The numbers of patients remaining at probability of deterioration are given alongside the X-axis denoting time in years. Tick marks indicate censored patients at the point of their last known follow-up. FSIQ: Full-Scale Intelligence Quotient; WMI : Working Memory Index, PSI: Processing Speed Index.

both mental reserve capacity and vulnerability and is the most critical determinant of the postoperative cognitive outcome [39]. Here, we did not observe any correlation between baseline FSIQ and neurocognitive outcome after epilepsy surgery probably because of the IQ distribution in our cohort. Indeed, 53/81 (65%) patients were below-average children and only 5/81 (6%) were above-average children. We noted that an intellectual improvement could occur regardless of the baseline intellectual abilities. From the baseline FSIQ, the postoperative improvement would correspond to normalization for some and would reduce the gap with the age group for others. In this way, the functional consequences of the postoperative intellectual outcome, meaning academic achievement, independence, and quality of life may be linked to the baseline performances.

These findings underline the importance of surgical management of patients with pharmacoresistant epilepsy as early as possible, as well as appropriate and timely neuropsychological assessment with serial follow-up.

4.2.2. After surgery

Entirely in line with recent literature [5], we show that, for most intellectual domains, postoperative improvement is more frequent than decline. Overall, 45 of the 81 (55%) recruited patients were improved on at least one of the five domains tested using standardized intellectual assessments after surgery. As Puka et al. reported an independent association between improved IQ and some academic domains, we could hypothesize that a change greater than + 10 points on one or many intellectual domains could have real academic achievement consequences [37].

The strongest predictors of intellectual improvement are epilepsy localization and seizure outcome. The degree of their impact on intellectual outcome is equivalent (Table 3). Our results agree with previous studies that viewed seizure freedom as the most important prognostic factor of intellectual development after surgery [5, 7, 10, 12, 14–16, 41].

- Neurocognitive Improvement

We further demonstrate that neurocognitive improvement after epilepsy surgery could be a long process that can be delayed for up to 5 years after surgery in seizure-free patients. Improvement occurred more frequently in the TLE group than in the ETLE group. Of the improved patients, 35 (78%) had TLE, and 37 (82%) were seizure-free. After epilepsy surgery, different intellectual domains such as working memory, processing speed, and perceptual reasoning tasks could be improved, but verbal tasks were only rarely improved, regardless of epilepsy lateralization. Here, we demonstrate that cognition is not uniformly improved by resection of the epileptogenic zone and achievement of seizure freedom. As proposed by Puka et al. [37], the presence of epileptic activity before seizure control is reached may represent an abnormal neural substrate that can hinder long-term language skills improvement, even in the absence of seizures. Furthermore, the word-finding difficulty sometimes observed immediately after left-sided cortectomy and challenges to fill gaps in lexical stock and semantic knowledge can explain the lack of significant improvement for VCI.

Cognitive changes following temporal lobectomy are better defined than cognitive effects of extratemporal resections [7, 36, 42, 43]. Our results showed that, even if prognosis in terms of seizure and intellectual outcomes tend to be less good in ETLE than TLE, an intellectual improvement can occur for those who become seizure-free [7, 36].

We observed here that factors such as the number of ASM impacted postoperative intellectual improvement through our univariate analysis. This finding confirms the well-known cognitive side-effects of ASM that are classically described, particularly in children [5] and how the postoperative reduction or complete withdrawal predicts an IQ gain [44]. However, compared to the impact of seizure freedom and epilepsy location, the ASM number and the ASM decrease had no significant effect on intellectual improvement in our study. This can be explained by

the very high percentage of patients with ASM monotherapy in our cohort (41% pre and postoperatively) compared to the literature.

- Neurocognitive decline

Conversely, 33% of patients experienced a decline after epilepsy surgery. Deterioration occurred more frequently in patients who were not seizure-free (57%) and in those with ETLE (63%). This intellectual impairment occurred in the first two years after the surgery, so it had a shorter time course than intellectual improvement. The observed intellectual decline involved only the FSIQ and the WMI. This data confirms the global impairment of intellectual functions induced by active epilepsy. Working memory, whose maturation continues into young adulthood, appears to be at risk of additional impairments corresponding to an increase in the gap with healthy peers of the same age. The decrease observed from the earliest post-surgery assessments is probably a continuation of the trend to decline observed before surgery. The same dynamic was observed for another specific cognitive function, verbal memory, in a six-year follow-up study of 85 patients with left-sided TLE. The patients had serial assessments over six years, and their verbal memory performance continued to decline for two years following left temporal lobe resection. It then stabilized over the next four years [45, 46].

4.3. Limitations

In this study, due to its retrospective design, the evaluation of specific functions (language, memory, and executive functions) using different specific psychometric tests was not systematic. We only analyzed the different cognitive domains of Wechsler's intelligence scales. The characterization of learning and memory deficits are essential for the patient's outcome in term of academic achievement and everyday functioning. Studies show that regardless of the temporal lobe epilepsy (TLE) side, verbal memory may be most affected in pediatric patients with TLE [47], and those who scored better at baseline are also at greater risk of loss [39]. These data may modulate cognitive outcomes in our TLE patients. A post-operative evaluation of these specific functions' evolution would have allowed us to better define the cognitive outcome after epilepsy surgery. Furthermore, the extratemporal categorization is often used in pediatric cohorts; it is however a heterogeneous group in terms of etiology, localization and extent of epilepsy, and as such may represent a bias in our analysis.

To minimize the risk that the increase in IQ might affect practice, we used a higher cut-off than those specified in the manuals. However, the use of RCI (Reliable Change Index) might allow a better control of the statistical properties of the test and, in particular, the reliability (better accounted for poor test-retest reliability, error measurement, practice effects, and regression towards the mean) [48].

Moreover, a prospective longitudinal study with a control group and matched patients according to epilepsy characteristics and IQ profile, should be conducted to further describe the cognitive changes over time observed after epilepsy surgery.

5. Conclusion

Our study is the first highlighting the time-course of IQ evolution before and after the epilepsy surgery. This longitudinal study revealed more frequent postoperative intellectual improvements than declines in patients with refractory epilepsy. Patients with TLE who achieved seizure freedom after surgery were the leading candidates for achieving intellectual improvement. Our study provides a dynamic view of intellectual outcome after epilepsy surgery. This specific timeline allows the neurologist to better guide the patient and his caregiver. An early pre-surgical evaluation for all children and adolescents with refractory epilepsy should be considered an emergency to avoid decline, accelerate the improvement of their intellectual functions, and ultimately may

improve their quality of life. Pre- and postsurgical cognitive assessments are essential given the difficulties in defining the developmental trajectory in children with epilepsy.

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